

CUSTOMER RETENTION AND CHURN ANALYSIS

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TOOLS USED:

- PYTHON
- PANDAS
- MATPLOTLIB
- VS CODE

Introduction:

Customer churn refers to the phenomenon where customers stop using a company's products or services over a period of time. In highly competitive industries such as telecommunications, customer retention is essential because acquiring new customers is often more expensive than retaining existing ones. Understanding why customers leave helps organizations improve customer satisfaction, reduce revenue loss, and develop effective retention strategies.

This project focuses on analyzing customer churn using the **IBM Telco Customer Churn** dataset. The dataset contains information about customer demographics, services subscribed, contract types, payment methods, monthly charges, tenure, and churn status. By performing data cleaning and exploratory data analysis (EDA), the project aims to identify patterns and relationships between customer characteristics and their likelihood of leaving the company.

The analysis was carried out using **Python**, along with libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**. Various visualizations were created to explore customer behavior, identify key factors influencing churn, and generate meaningful business insights. The findings from this analysis can help organizations make data-driven decisions to improve customer retention, enhance service quality, and design strategies that minimize customer churn.

Project Objective:

The primary objective of this project is to analyze customer churn using the IBM Telco Customer Churn dataset and identify the key factors influencing customer retention. By understanding customer behavior and service usage patterns, the analysis aims to provide meaningful insights that can help businesses reduce customer churn and improve customer satisfaction.

The specific objectives of this project are:

- To understand the structure and characteristics of the customer churn dataset.
- To clean and preprocess the dataset by handling missing values and ensuring data quality.
- To perform Exploratory Data Analysis (EDA) using statistical methods and visualizations.
- To identify the relationship between customer churn and factors such as contract type, internet service, payment method, tenure, and monthly charges.
- To generate business insights that can support data-driven decision-making.
- To provide practical recommendations for improving customer retention and reducing customer churn.

Dataset Description:

The **IBM Telco Customer Churn Dataset** is a widely used dataset for analyzing customer retention in the telecommunications industry. It contains information about **7,043 customers** with **21 features**, including customer demographics, subscribed services, contract details, billing information, and churn status. The target variable, **Churn**, indicates whether a customer has left the company's services or continues to use them. After data preprocessing, **11 records with missing values** were removed, resulting in a cleaned dataset of **7,032 records**. This dataset helps identify the key factors influencing customer churn and supports data-driven business decisions to improve customer retention.

Key Features of the Dataset

CustomerID: A unique identification number assigned to each customer.

Gender: Indicates whether the customer is male or female.

SeniorCitizen: Specifies whether the customer is a senior citizen (1 = Yes, 0 = No).

Partner: Indicates whether the customer has a spouse or partner.

Dependents: Shows whether the customer has dependents such as children or family members.

Tenure: Represents the number of months the customer has remained with the company.

PhoneService: Indicates whether the customer has subscribed to the phone service.

InternetService: Specifies the type of internet service used by the customer, such as DSL, Fiber Optic, or No Internet Service.

Contract: Represents the type of contract chosen by the customer, such as Month-to-Month, One Year, or Two Year.

PaperlessBilling: Indicates whether the customer receives paperless billing.

PaymentMethod: Specifies the method used by the customer to pay monthly bills.

MonthlyCharges: Represents the amount charged to the customer every month for the subscribed services.

TotalCharges: Indicates the total amount paid by the customer throughout their subscription period.

Churn: The target variable that indicates whether the customer has left the company's service (Yes) or continues to use it (No).

Data Cleaning & Preprocessing:

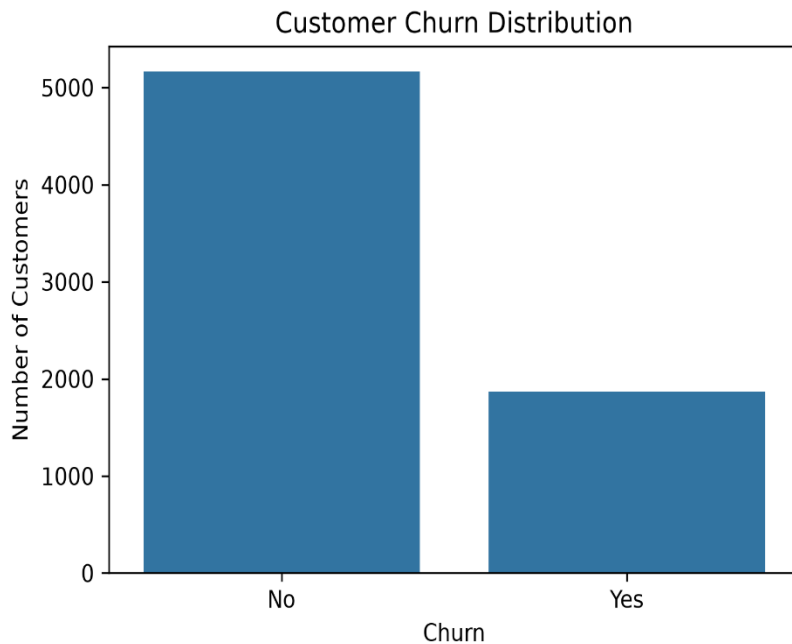
Before performing the analysis, the dataset was preprocessed to ensure data quality and consistency. The dataset was first examined for missing values, duplicate records, and data types. The TotalCharges column was converted from text to a numeric data type using the `pd.to_numeric()` function. Missing values present in the TotalCharges column were identified and removed using the `dropna()` function. Duplicate records were also checked, and no duplicate entries were found. After completing the preprocessing steps, the dataset contained **7,032 clean records**, making it suitable for exploratory data analysis and visualization.

Data Cleaning Steps

- Loaded the dataset into Jupyter Notebook using Pandas.
- Checked the dataset shape, column names, and data types.
- Identified missing values using `isnull().sum()`.
- Converted the TotalCharges column into numeric format.
- Removed rows containing missing values.
- Verified that no duplicate records were present.
- Prepared the cleaned dataset for visualization and analysis.

Analysis and Charts:

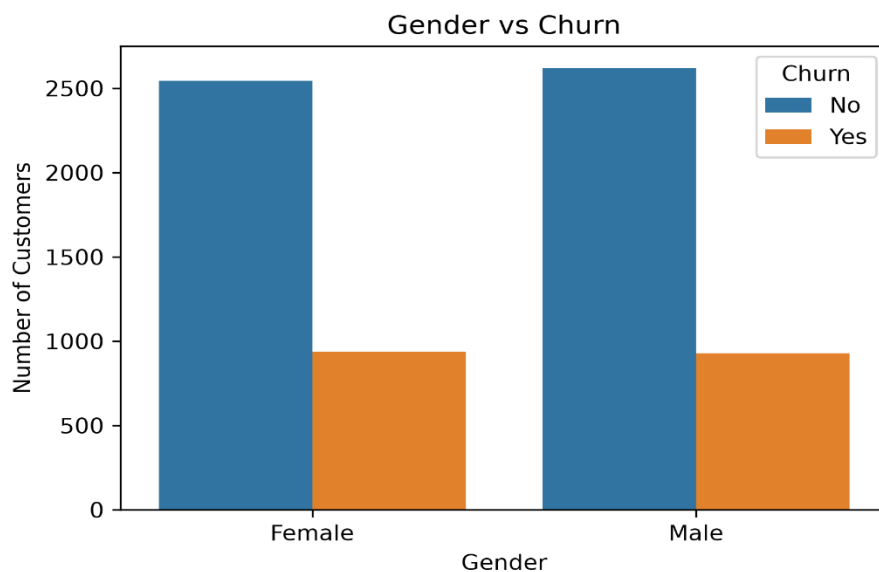
Chart 1: Customer Churn Distribution



Analysis:

- The chart shows the number of customers who stayed and those who left the company.
- Most customers continued using the company's services, while fewer customers churned.
- The company has a higher customer retention rate than churn rate.
- Reducing the churned customer group can further improve business growth.

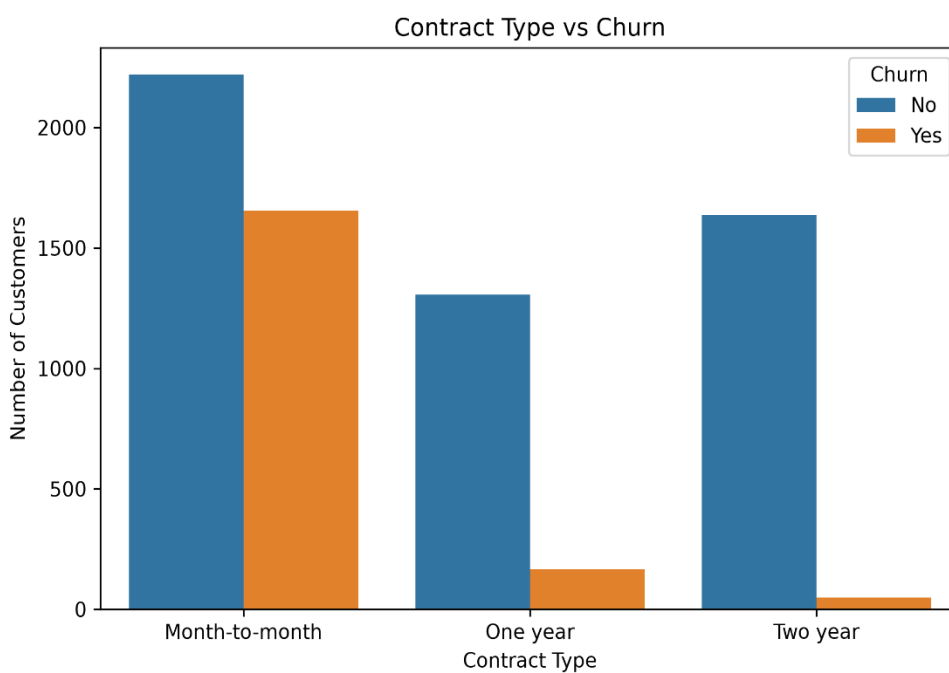
Chart 2: Gender vs Customer Churn



Analysis:

- The chart compares customer churn based on gender.
- Male and female customers show similar churn patterns.
- Gender does not appear to have a significant impact on customer churn.
- The company should focus more on service-related factors than gender.

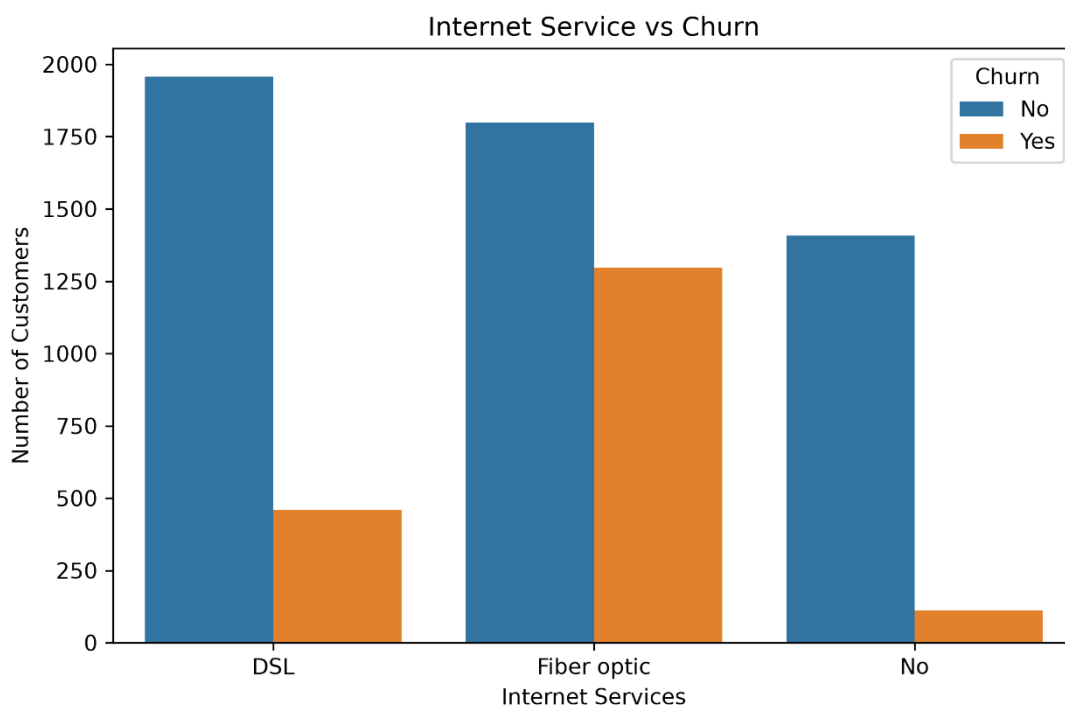
Chart 3: Contract Type vs Customer Churn



Analysis:

- The chart shows customer churn for different contract types.
- Customers with Month-to-Month contracts have the highest churn rate.
- One-Year and Two-Year contract customers are more likely to stay.
- Encouraging long-term contracts can improve customer retention.

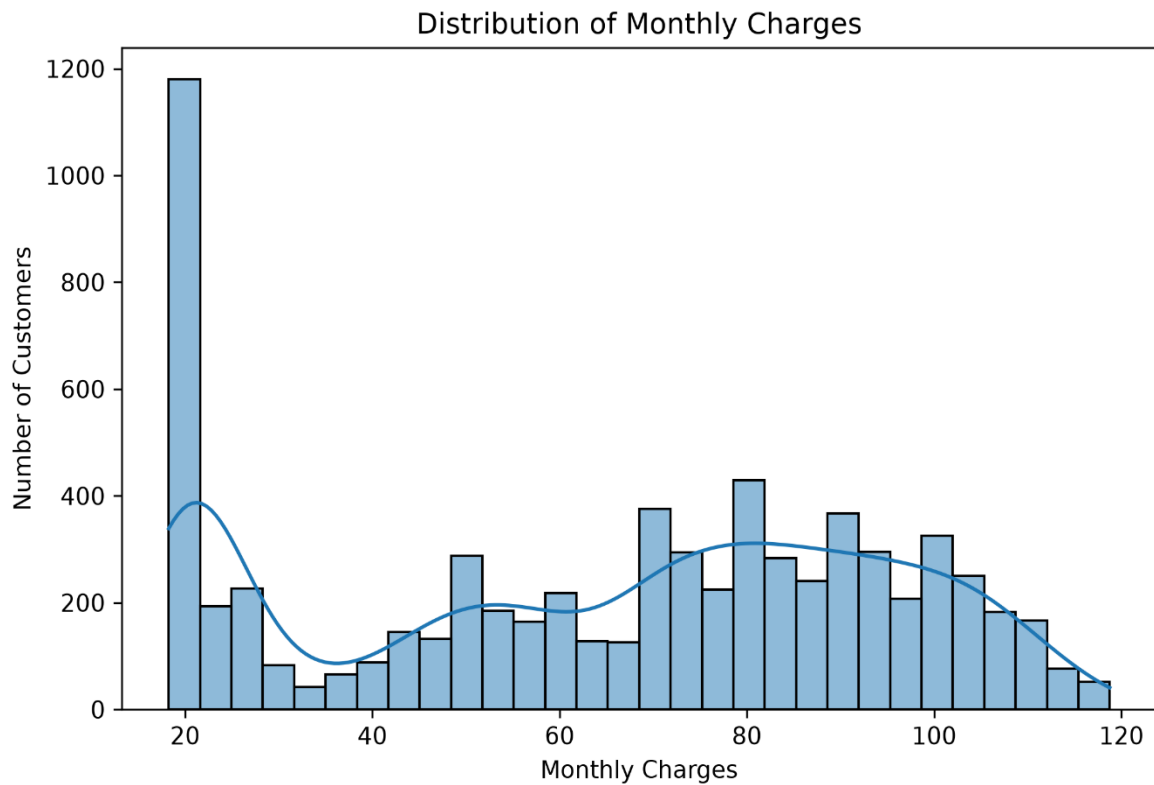
Chart 4: Internet Service vs Customer Churn



Analysis:

- The chart compares churn among different internet service types.
- Fiber Optic customers have a higher churn rate than DSL customers.
- Internet service quality or pricing may influence customer decisions.
- Improving Fiber Optic services can help reduce customer churn.

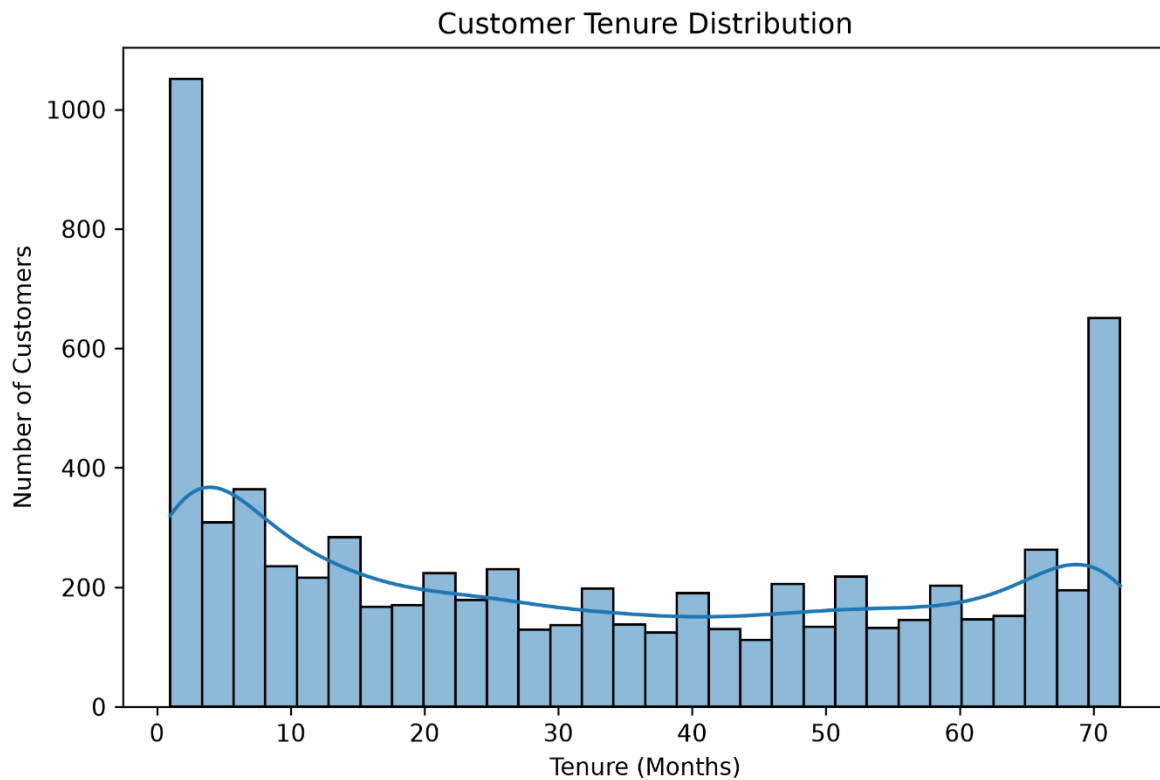
Chart 5: Monthly Charges Distribution



Analysis:

- The chart shows how monthly charges are distributed among customers.
- Customers pay different monthly charges based on their subscribed services.
- Higher monthly charges may influence some customers to leave.
- Reviewing pricing plans can help improve customer satisfaction.

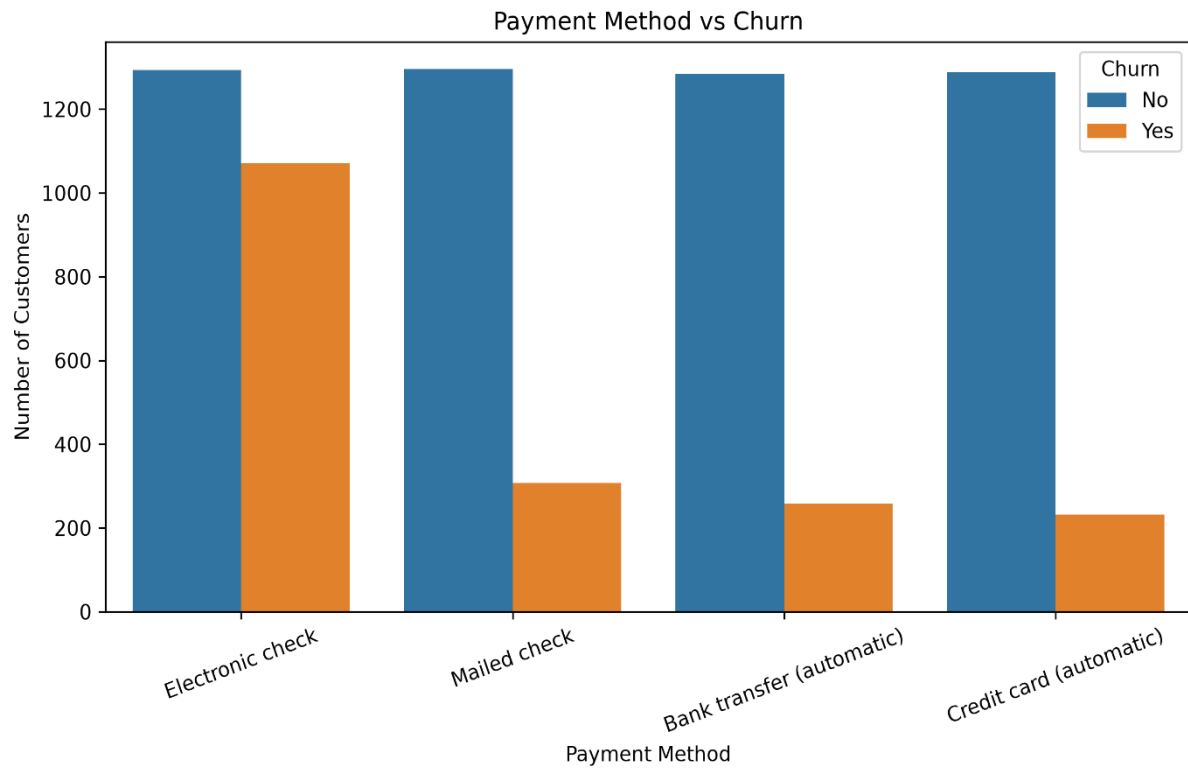
Chart 6: Customer Tenure Distribution



Analysis:

- The chart shows how long customers have stayed with the company.
- Many customers have shorter tenures, while fewer have very long tenures.
- Customers with longer tenure tend to be more loyal.
- Improving the experience of new customers can increase long-term retention.

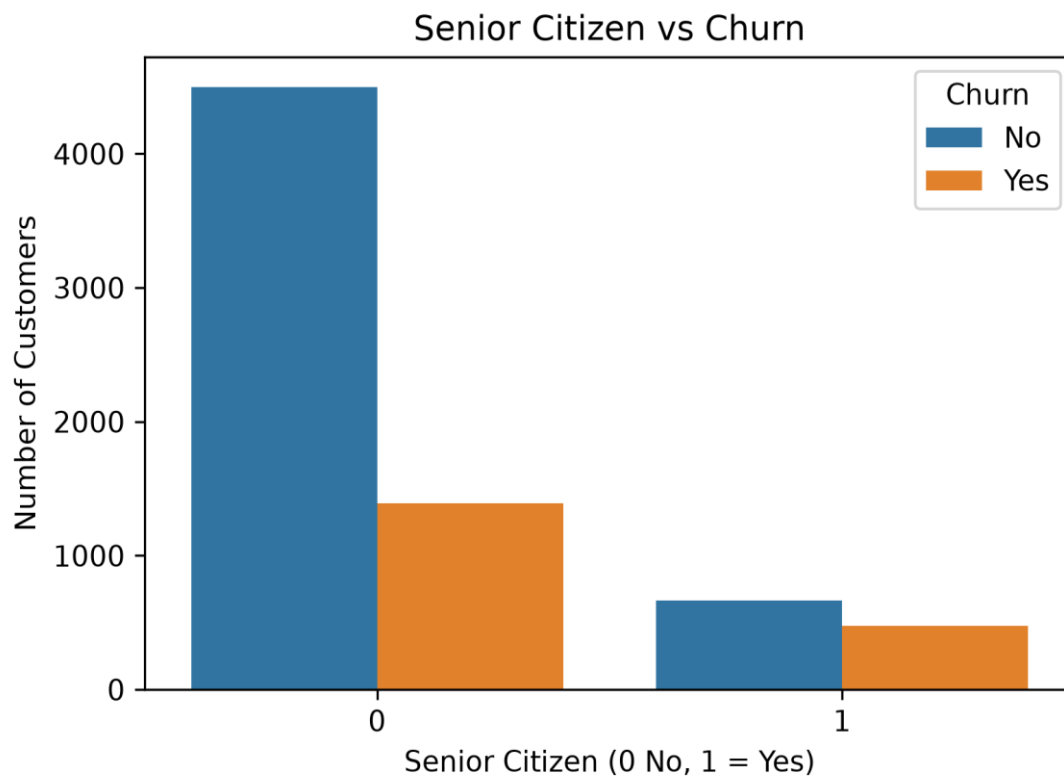
Chart 7: Payment Method vs Customer Churn



Analysis:

- The chart compares customer churn across different payment methods.
- Customers using Electronic Check show higher churn.
- Payment preferences may influence customer experience and retention.
- Promoting convenient payment methods may help reduce churn.

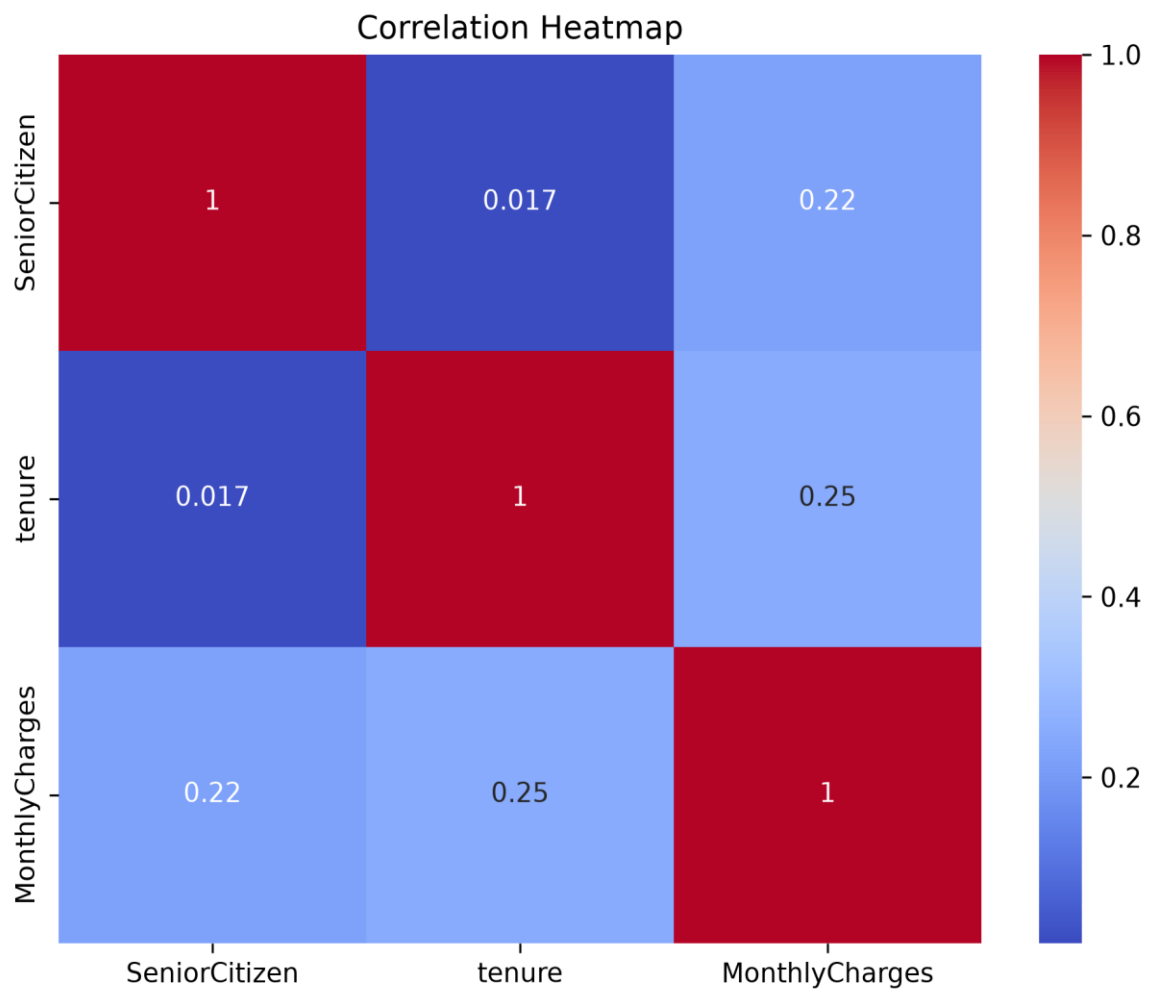
Chart 8: Senior Citizen vs Customer Churn



Analysis:

- The chart compares churn between senior citizens and non-senior citizens.
- Churn rates differ between the two customer groups.
- Age may influence customer behavior and service expectations.
- Providing personalized support for senior citizens may improve retention.

Chart 9: Correlation Heatmap



Analysis:

- The heatmap shows the relationship between numerical features in the dataset.
- Most features have weak to moderate correlations with each other.
- No single numerical feature strongly determines customer churn.
- Customer churn is influenced by multiple factors rather than one variable alone.

Key Insights:

1. **Customer retention is higher than customer churn**, indicating that the company has successfully retained a majority of its customers.
2. **Customers with Month-to-Month contracts are more likely to churn** compared to those with One-Year or Two-Year contracts, highlighting the importance of long-term commitments.
3. **Fiber Optic internet users show a relatively higher churn rate**, suggesting that service quality, pricing, or customer expectations may require improvement.
4. **Electronic Check is the payment method most commonly associated with churn**, indicating that payment preferences may influence customer satisfaction.
5. **Customers with longer tenure are generally more loyal**, while newer customers are at a higher risk of leaving the company.
6. **Monthly charges appear to influence customer behavior**, as customers with higher monthly bills may be more likely to discontinue the service.
7. **Gender has minimal impact on customer churn**, suggesting that demographic characteristics alone are not strong indicators of customer retention.
8. **Customer churn is influenced by multiple factors**, including contract type, tenure, internet service, payment method, and billing patterns, rather than a single variable.
9. **Exploratory Data Analysis (EDA) effectively identified customer behavior patterns**, helping to understand the major factors contributing to customer churn.
10. **The insights obtained from this analysis can support data-driven business decisions** aimed at improving customer retention, enhancing service quality, and reducing customer churn.

Recommendations:

- **Promote long-term contracts** by offering discounts or special benefits to customers choosing one-year or two-year plans.
- **Improve the quality of Fiber Optic services** by addressing customer complaints and enhancing network performance.
- **Introduce customer loyalty programs** to reward long-term customers and encourage continued subscriptions.
- **Enhance support for new customers** through better onboarding and regular follow-ups during the initial months.
- **Encourage secure and convenient payment methods**, such as automatic bank transfers or credit card payments, to improve customer experience.
- **Regularly monitor customer feedback and satisfaction** to identify issues early and take proactive measures to reduce churn.
- **Use data analytics to identify high-risk customers** and provide personalized offers or support to improve retention.

Conclusion:

This project successfully analyzed customer churn using the **IBM Telco Customer Churn Dataset** through data cleaning and Exploratory Data Analysis (EDA). The analysis revealed that factors such as **contract type, internet service, payment method, monthly charges, and customer tenure** have a significant influence on customer churn. Visualizations helped identify important patterns in customer behavior and provided valuable insights into the reasons behind customer retention and customer loss. These findings can help businesses make informed decisions, improve customer satisfaction, implement effective retention strategies, and reduce customer churn. Overall, this project demonstrates the importance of data analysis in understanding customer behavior and supporting better business decision-making.